

Transvergent: Understanding Tree

Understanding The Understanding Tree

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Introduction.....	3
How It Works	3
The Primary Scene.....	5
The Node Scene	6
Exporting	6
Importing.....	6

Introduction

The Understanding Tree demo (further referred to as “UTree”) was written in **Godot**. Godot uses a language highly similar to Python called **GDScript**. Godot is an open source, community driven platform. By developing a demo through this program, we preserve the accessibility of the demo well beyond the lifespan of a web application.

The UTree is a diagrammatic representation of problem solving. It allows the user to uniquely visualize a problem and its solutions.

How It Works

(programmatically)

The UTree system uses **nodes which contain a problem and a solution**. The “nodes” are of minimal importance to the user, but it serves a major purpose in the backend. The UTree uses a starting node, off of which subsequent child nodes stem from. All nodes lead back to the original starting node. Subsequent child nodes can be children either of the original node itself or a problem/solution.

The user does not edit nodes. The user edits problems and solutions through the usage of **4 action buttons: delete, expand, diverge, and counter argument**.

1. Delete: Deletes the problem/solution, either making the problem/solution invisible, or deletes the entire node *along with all children*
2. Expand: Creates a new node at the *same angle* of the problem/solution selected, as a child of the *selected problem/solution* and of the same type.
3. Diverge: Creates a new node at a *vertical angle* of the problem/solution selected, as a child of the *selected node* and of the same type.
4. Counterargument: Creates a new node at a *horizontal angle* of the problem/solution selected, as a child of the *selected node* and of the same type.
OR, if there is no problem/solution oppositional to the selected problem/solution, that oppositional problem/solution becomes visible

How To Access

Godot, being an open source project, is free and available to all. Godot can be downloaded from this URL: <https://godotengine.org/>.

Godot uses a language highly similar to Python called **GDScript**. GDScript changes across version to version, but the version the code was written in should remain functional across OS frameworks. The code for this project was written in Godot version 4.3. Archived versions of Godot can be accessed through this URL: <https://godotengine.org/download/archive/>.

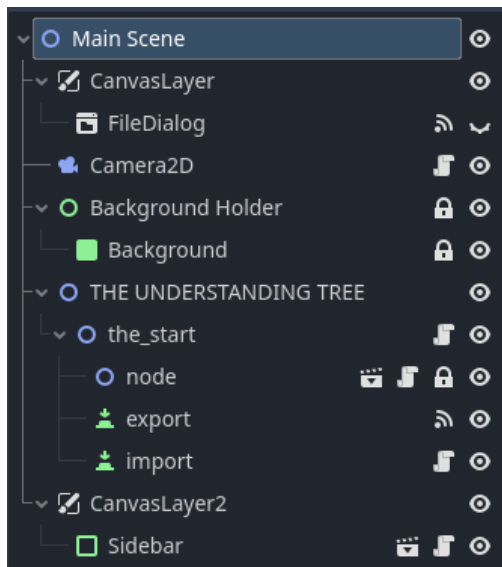


All files from our production of the Understanding Tree are available through GitHub at this link: <https://github.com/eqgy/understandingtree>. The iteration can be found under “application iterations/iteration 1”.

The following steps should be used to open the project:

- Download Godot from the official Godot Engine website: <https://godotengine.org/>.
- Open and extract the folder
- Run the .exe file (ideally the non-console one)
- As of writing, in the top left there should be an import button. Click it.
- Find the file “application iterations/iteration 1/project.godot”.
- The project should be imported successfully. To run the project, press “F6” or click the play button in the top right.

The Main Scene



The main scene contains a few objects:

- CanvasLayer
- Camera2D
- Background Holder
- “THE UNDERSTANDING TREE”
- CanvasLayer2

CanvasLayer: Holds a file dialog that appears upon import being pressed

Camera2D: Is the viewport through which the player views the

Background Holder: Holds a colored background object

“THE UNDERSTANDING TREE”: Contains no script itself. Is just a divider. Contains “the_start”, which handles **everything** within the tree. “the_start” contains the export and import buttons, as well as the original node.

CanvasLayer2: Contains the sidebar scene.

The Node Scene

Exporting

Importing